Freight Rail Carload Forecasting: GAM Model — Symbolic Specification

`GAMSpline` class • `freight_pipeline_v2.py` • Three targets: *originated*, *received*, *total*

1. Notation

Symbol	Meaning
$t \in \{0, 1, \ldots, T-1\}$	Integer week index (zero-based)
$y_t \in \mathbb{R}_{\geq 0}$	Aggregate weekly carloads (target variable)
$\mathcal{C}$	Set of railway companies, $\lvert\mathcal{C}\rvert = C$
$\mathcal{K}$	Set of commodity codes, $\lvert\mathcal{K}\rvert = K$
$w_{c,t} \in [0, 1]$	Share of total carloads attributable to company c at week t
$v_{k,t} \in [0, 1]$	Share of total carloads attributable to commodity k at week t
$\tau \in \mathbb{N}$	Number of interior trend knots
$\kappa \in \mathbb{N}$	Number of interior entity knots (shared across all entities)
$H \in \mathbb{N}$	Number of Fourier harmonics
$\lambda \in \mathbb{R}_{>0}$	Ridge (L2) regularisation strength
$\phi(\cdot)$	Cubic B-spline basis map
$S = 365.25/7$	Annual seasonal period (weeks)

2. Master Structural Equation

Let y_t denote the aggregate weekly carload total for a given target. The model decomposes the signal additively:

$$\boxed{y_t \;=\; f_{\text{trend}}(t) + \sum_{c \in \mathcal{C}} f_c(t)\, w_{c,t} + \sum_{k \in \mathcal{K}} f_k(t)\, v_{k,t} + f_{\text{seas}}(t) + f_{\text{cal}}(t) + \varepsilon_t} \tag{1}$$

where $\varepsilon_t \overset{\text{iid}}{\sim} \mathcal{N}(0, \sigma^2)$ is the idiosyncratic error, and every smooth function $f_\bullet$ is estimated jointly via Ridge-penalised regression on a standardised design matrix (Section 6).

3. Component Functions

3.1. Global Trend Spline — f_{trend}

The global trend is captured by a cubic B-spline with τ interior knots placed at the quantiles of the observed week indices:

$$f_{\text{trend}}(t) \;=\; \sum_{j=1}^{\tau+4} \beta_j^{\text{tr}}\, B_j^{(3)}(t;\, \boldsymbol{\xi}^{\text{tr}}) \tag{2}$$

where $B_j^{(3)}(\cdot\,;\, \boldsymbol{\xi}^{\text{tr}})$ is the j-th cubic (*degree-3*) B-spline basis function defined over the knot sequence

$$\boldsymbol{\xi}^{\text{tr}} = \big(\underbrace{t_0,\, t_0,\, t_0,\, t_0}_{\text{boundary }(\times 4)},\; \xi_1^{\text{tr}}, \ldots, \xi_\tau^{\text{tr}},\; \underbrace{t_{T-1},\, t_{T-1},\, t_{T-1},\, t_{T-1}}_{\text{boundary }(\times 4)}\big)$$

with the τ interior knots $\xi_1^{\text{tr}} < \cdots < \xi_\tau^{\text{tr}}$ located at the $\left\{\frac{i}{\tau+1}\right\}_{i=1}^{\tau}$ empirical quantiles of $\{t\}_{t=0}^{T-1}$.

The trend spline yields $\tau + 4$ basis columns (three fewer than the knot sequence length, following standard B-spline construction).

3.2. Company Entity Splines — f_c

Each company $c \in \mathcal{C}$ receives its own smooth time function $f_c : \mathbb{R} \to \mathbb{R}$, parameterised by a separate B-spline with κ interior knots:

$$f_c(t) \;=\; \sum_{j=1}^{\kappa+4} \beta_j^{(c)}\, B_j^{(3)}(t;\, \boldsymbol{\xi}^{(c)}) \tag{3}$$

The entity knots $\boldsymbol{\xi}^{(c)}$ share the same quantile placement scheme as the trend spline but use $\kappa < \tau$ interior knots (shallower flexibility).

The weighted contribution to the aggregate signal is $f_c(t)\, w_{c,t}$, where $w_{c,t}$ is the fraction of the weekly total attributable to company c (soft indicator; $\sum_c w_{c,t} = 1$).

3.3. Commodity Entity Splines — f_k

Identically structured to company splines:

$$f_k(t) \;=\; \sum_{j=1}^{\kappa+4} \beta_j^{(k)}\, B_j^{(3)}(t;\, \boldsymbol{\xi}^{(k)}) \tag{4}$$

with commodity weight $v_{k,t}$ (fraction of weekly total in code k; $\sum_k v_{k,t} = 1$).

3.4. Fourier Seasonality — f_{seas}

Annual seasonality is modelled as a finite Fourier series of order H:

$$f_{\text{seas}}(t) \;=\; \sum_{h=1}^{H} \left[\alpha_h \sin\!\left(\frac{2\pi h\, t}{S}\right) + \gamma_h \cos\!\left(\frac{2\pi h\, t}{S}\right)\right] \tag{5}$$

where $S = 365.25/7 \approx 52.18$ weeks and the $2H$ Fourier coefficients $\{\alpha_h, \gamma_h\}_{h=1}^{H}$ are estimated jointly with all other model parameters.

3.5. Calendar Effects — f_{cal}

A set of five binary calendar indicators captures systematic departures from the underlying seasonal pattern:

$$f_{\text{cal}}(t) \;=\; \sum_{\ell=1}^{5} \delta_\ell \, \mathbf{1}_{A_\ell}(t) \tag{6}$$

where the indicator sets are

$$\begin{aligned}
A_1 &= \{t : \text{quarter}(t) = 1\} && \text{(Q1)} \\
A_2 &= \{t : \text{quarter}(t) = 2\} && \text{(Q2)} \\
A_3 &= \{t : \text{quarter}(t) = 3\} && \text{(Q3)} \\
A_4 &= \{t : \text{quarter}(t) = 4\} && \text{(Q4)} \\
A_5 &= \{t : \text{woy}(t) \in \{1, 52, 53\}\} && \text{(holiday-boundary weeks)}
\end{aligned}$$

and $\text{woy}(t)$ denotes the ISO week-of-year for week t.

4. Design Matrix

Collecting all basis functions, the design matrix $\mathbf{X} \in \mathbb{R}^{T \times p}$ is the horizontal concatenation

$$\begin{aligned}
\mathbf{X} = \Big[\underbrace{\mathbf{B}^{\text{tr}}}_{T \times (\tau+4)} \;\Big|\; & \underbrace{\mathbf{B}^{(c_1)}\text{diag}(\mathbf{w}_{c_1}),\; \ldots,\; \mathbf{B}^{(c_C)}\text{diag}(\mathbf{w}_{c_C})}_{T \times C(\kappa+4)} \\
& \Big|\; \underbrace{\mathbf{B}^{(k_1)}\text{diag}(\mathbf{v}_{k_1}),\; \ldots,\; \mathbf{B}^{(k_K)}\text{diag}(\mathbf{v}_{k_K})}_{T \times K(\kappa+4)} \;\Big|\; \underbrace{\mathbf{F}}_{T \times 2H} \;\Big|\; \underbrace{\mathbf{D}}_{T \times 5} \Big]
\end{aligned} \tag{7}$$

where:

- $\mathbf{B}^{\text{tr}} \in \mathbb{R}^{T \times (\tau+4)}$ is the trend B-spline basis matrix evaluated at $t = 0, \ldots, T-1$.
- $\mathbf{B}^{(c)}, \mathbf{B}^{(k)} \in \mathbb{R}^{T \times (\kappa+4)}$ are entity-specific B-spline bases; each column is multiplied element-wise by the corresponding weight vector $\mathbf{w}_c$ or $\mathbf{v}_k$.
- $\mathbf{F} \in \mathbb{R}^{T \times 2H}$ contains the stacked sine and cosine Fourier columns.
- $\mathbf{D} \in \mathbb{R}^{T \times 5}$ contains the five binary calendar indicators.

The total parameter count is

$$p \;=\; (\tau + 4) + C(\kappa + 4) + K(\kappa + 4) + 2H + 5 \tag{8}$$

5. Feature Standardisation

Prior to estimation, every column of $\mathbf{X}$ is standardised using the training-set mean μ_j and standard deviation s_j:

$$\tilde{x}_{t,j} \;=\; \frac{x_{t,j} - \mu_j}{s_j}, \qquad j = 1, \ldots, p \tag{9}$$

This yields the standardised matrix $\tilde{\mathbf{X}}$, which is passed directly to the Ridge estimator.

6. Estimation

6.1. Ridge (L2) Objective

All parameters are estimated jointly by minimising the Ridge-penalised least-squares criterion:

$$\hat{\boldsymbol{\beta}} \;=\; \underset{\boldsymbol{\beta} \in \mathbb{R}^p}{\arg\min} \left\{ \left\| \mathbf{y} - \tilde{\mathbf{X}}\boldsymbol{\beta} \right\|_2^2 \;+\; \lambda \left\| \boldsymbol{\beta} \right\|_2^2 \right\} \tag{10}$$

where $\mathbf{y} = (y_0, \ldots, y_{T-1})^\top$ is the target vector and $\lambda > 0$ is the Ridge regularisation strength (selected by cross-validation; see Section 8).

6.2. Closed-Form Solution

The Ridge estimator has the closed-form solution

$$\hat{\boldsymbol{\beta}} \;=\; \left(\tilde{\mathbf{X}}^\top \tilde{\mathbf{X}} + \lambda \mathbf{I}_p\right)^{-1} \tilde{\mathbf{X}}^\top \mathbf{y} \tag{11}$$

The intercept $\beta_0 \in \mathbb{R}$ is estimated separately (without regularisation) as the mean of the centred residuals.

6.3. Fitted Values and Prediction

The in-sample fitted values are

$$\hat{y}_t \;=\; \beta_0 + \tilde{\mathbf{x}}_t^\top \hat{\boldsymbol{\beta}} \;=\; \beta_0 + \hat{f}_{\text{trend}}(t) + \sum_c \hat{f}_c(t)\, w_{c,t} + \sum_k \hat{f}_k(t)\, v_{k,t} + \hat{f}_{\text{seas}}(t) + \hat{f}_{\text{cal}}(t) \tag{12}$$

For h-step-ahead prediction at future week $t^* = T - 1 + h$, the entity weights w_{c,t^*} and v_{k,t^*} are set to their last observed values (naive forward carry), and the forecast is clipped to be non-negative:

$$\hat{y}_{t^*} \;=\; \max\!\left(0,\; \beta_0 + \tilde{\mathbf{x}}_{t^*}^\top \hat{\boldsymbol{\beta}}\right) \tag{13}$$

7. Additive Decomposition

Because the model is purely additive and linear in the coefficients, fitted components can be recovered exactly post-estimation:

$$\hat{f}_{\text{trend}}(t) = \tilde{\mathbf{b}}^{\text{tr}}(t)^\top \hat{\boldsymbol{\beta}}^{\text{tr}} \tag{14}$$

$$\hat{f}_c(t) = w_{c,t}\, \tilde{\mathbf{b}}^{(c)}(t)^\top \hat{\boldsymbol{\beta}}^{(c)} \tag{15}$$

$$\hat{f}_k(t) = v_{k,t}\, \tilde{\mathbf{b}}^{(k)}(t)^\top \hat{\boldsymbol{\beta}}^{(k)} \tag{16}$$

$$\hat{f}_{\text{seas}}(t) = \tilde{\mathbf{f}}(t)^\top \hat{\boldsymbol{\alpha}} \tag{17}$$

$$\hat{f}_{\text{cal}}(t) = \mathbf{d}(t)^\top \hat{\boldsymbol{\delta}} \tag{18}$$

where $\tilde{\mathbf{b}}(\cdot)$, $\tilde{\mathbf{f}}(\cdot)$, and $\mathbf{d}(\cdot)$ denote the standardised rows of the respective sub-matrices of $\tilde{\mathbf{X}}$, and $\hat{\boldsymbol{\alpha}}$, $\hat{\boldsymbol{\delta}}$ are the corresponding sub-vectors of $\hat{\boldsymbol{\beta}}$.

8. Hyperparameter Selection

The two key hyperparameters — Ridge strength λ and interior knot count τ — are chosen by minimising the mean cross-validated RMSE over a 5×2 repeated time-series cross-validation scheme:

8.1. CV Split Construction

For training sample size T_{tr}, CV fold r uses a training prefix of length $\lfloor T_{\text{tr}} \cdot \rho_r \rfloor$ and a contiguous test window of $\lfloor T_{\text{tr}} \cdot \phi \rfloor$ observations immediately following, where

$$\rho_r \sim \text{Uniform}(0.45,\, 0.85) \quad \text{(jittered base fraction)} \tag{19}$$

$$\phi = 0.12 \quad \text{(fixed test fraction)} \tag{20}$$

The jitter prevents all folds from being anchored at the same cut-point, yielding $R = N_{\text{rep}} \times K = 5 \times 2 = 10$ folds total.

8.2. Selection Criterion

Let $\hat{y}_t^{(r)}$ denote the out-of-sample prediction for week t in fold r. The selection criterion is

$$(\hat{\lambda},\, \hat{\tau}) \;=\; \underset{\lambda,\,\tau}{\arg\min}\; \frac{1}{R} \sum_{r=1}^{R} \sqrt{\frac{1}{|\mathcal{T}_r|} \sum_{t \in \mathcal{T}_r} \big(y_t - \hat{y}_t^{(r)}\big)^2} \tag{21}$$

where $\mathcal{T}_r$ is the test index set for fold r.

8.3. Search Grid

9. Model Identifiability and Regularisation

Collinearity. When C and K are large, the company and commodity spline columns may be nearly collinear with the global trend spline. Ridge regression mitigates this by shrinking all coefficients towards zero, trading a small amount of bias for a substantial reduction in variance.

Hyperparameter	Candidate values
λ (Ridge strength)	$\{0.1,\ 0.5,\ 1.0,\ 5.0,\ 10.0,\ 50.0\}$
τ (trend interior knots)	$\{8,\ 12,\ 15,\ 20,\ 25\}$
κ (entity interior knots)	Fixed at 8
H (Fourier harmonics)	Fixed at 12

Table 1: Hyperparameter search grid. $|\lambda| \times |\tau| = 30$ candidate configurations evaluated per target.

Interpretability of components. Because all features share a common intercept and are jointly standardised, the raw coefficients $\hat{\beta}_j$ reflect *standardised* effect sizes. The unstandardised (scale-native) effect size for feature j is recovered as $\hat{\beta}_j / s_j$.

Knot placement. Using *quantile* knot placement (rather than uniform spacing) concentrates flexibility where observations are dense, which is preferable for weekly freight data whose density is approximately uniform in calendar time but may be uneven in *week-number* space near data boundaries.

10. Evaluation Metrics

Let $\{y_t, \hat{y}_t\}_{t \in \mathcal{T}_{\text{eval}}}$ denote actual and predicted carloads on the evaluation set.

$$\text{RMSE} = \sqrt{\frac{1}{|\mathcal{T}|} \sum_{t \in \mathcal{T}} \big(y_t - \hat{y}_t\big)^2} \tag{22}$$

$$\text{MAE} = \frac{1}{|\mathcal{T}|} \sum_{t \in \mathcal{T}} \big|y_t - \hat{y}_t\big| \tag{23}$$

$$\text{MAPE} = \frac{100}{|\mathcal{T}|} \sum_{t \in \mathcal{T}} \frac{|y_t - \hat{y}_t|}{|y_t| + 1} \tag{24}$$

$$R^2 = 1 - \frac{\sum_t (y_t - \hat{y}_t)^2}{\sum_t (y_t - \bar{y})^2} \tag{25}$$

The $+1$ stabiliser in the MAPE denominator prevents division by zero for zero-valued actuals.

11. Summary of Model Dimensions

A. B-Spline Basis — Formal Definition

A cubic B-spline basis of order 4 (degree 3) over knot sequence $\Xi = (\xi_1 \leq \xi_2 \leq \cdots \leq \xi_m)$ is defined recursively via the Cox–de Boor recurrence:

Block	**Columns**	**Parameters**
Global trend spline	$\tau + 4$	$\tau + 4$
Company entity splines	$C\,(\kappa + 4)$	$C\,(\kappa + 4)$
Commodity entity splines	$K\,(\kappa + 4)$	$K\,(\kappa + 4)$
Fourier seasonality	$2H$	$2H$
Calendar indicators	5	5
Intercept	—	1
Total	p	$p + 1$

Table 2: Design matrix block structure. $p = (\tau + 4) + (C + K)(\kappa + 4) + 2H + 5$.

$$B_{j,1}(t;\Xi) = \mathbf{1}_{[\xi_j,\,\xi_{j+1})}(t) \tag{26}$$

$$B_{j,d}(t;\Xi) = \frac{t - \xi_j}{\xi_{j+d-1} - \xi_j}\, B_{j,\,d-1}(t;\Xi) + \frac{\xi_{j+d} - t}{\xi_{j+d} - \xi_{j+1}}\, B_{j+1,\,d-1}(t;\Xi) \tag{27}$$

with the convention $0/0 \coloneqq 0$. For degree $d = 4$ (cubic), each basis function has support on at most four consecutive knot spans and satisfies $\sum_j B_{j,4}(t;\Xi) = 1$ for all t in the interior.

The basis matrix for the trend component is therefore

$$\mathbf{B}^{\mathrm{tr}} \;=\; \begin{pmatrix} B_{1,4}(t_0;\boldsymbol{\xi}^{\mathrm{tr}}) & \cdots & B_{\tau+4,4}(t_0;\boldsymbol{\xi}^{\mathrm{tr}}) \\ \vdots & \ddots & \vdots \\ B_{1,4}(t_{T-1};\boldsymbol{\xi}^{\mathrm{tr}}) & \cdots & B_{\tau+4,4}(t_{T-1};\boldsymbol{\xi}^{\mathrm{tr}}) \end{pmatrix} \in \mathbb{R}^{T\times(\tau+4)} \tag{28}$$

B. Ridge Estimator — Degrees of Freedom

The effective degrees of freedom for a Ridge estimator are

$$\mathrm{df}(\lambda) \;=\; \mathrm{tr}\Big[\tilde{\mathbf{X}}\big(\tilde{\mathbf{X}}^{\top}\tilde{\mathbf{X}} + \lambda\mathbf{I}\big)^{-1}\tilde{\mathbf{X}}^{\top}\Big] \;=\; \sum_{j=1}^{p} \frac{d_j^2}{d_j^2 + \lambda} \tag{29}$$

where $d_1 \geq d_2 \geq \cdots \geq d_p \geq 0$ are the singular values of $\tilde{\mathbf{X}}$. As $\lambda \to 0$, $\mathrm{df} \to p$ (OLS); as $\lambda \to \infty$, $\mathrm{df} \to 0$ (full shrinkage).